

Multi-Prompting Scenario-based Movie Recommendation with Large Language Models: Real User Case Study

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Abstract

This paper investigates the potential of using large language models (LLMs) for personalized movie recommendations in an online field experiment. We assess the performance of LLM recommenders using a combination of between-subject prompts, historical consumption patterns, and within-subject recommendation scenarios. Analyzing conversation and survey data from 160 active users, we find that while LLMs excel in providing explainable recommendations, they lack in personalization, diversity, and user trust. Interestingly, personalized prompting techniques do not significantly affect user-perceived recommendation quality, while the number of movies a user has watched plays a more significant role. Furthermore, LLMs demonstrate a stronger ability to recommend lesser-known or niche movies. Through qualitative analysis, we identify key conversational patterns linked to positive and negative user interaction experiences and conclude that providing personal context and examples is crucial for obtaining high-quality recommendations from LLMs. These insights offer practical implications for improving LLM-based RecSys in real-world applications.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Information systems** → **Users and interactive retrieval**; **Recommender systems**; • **Computing methodologies** → **Natural language generation**.

Keywords

Large Language Model, Generative AI, Recommender System, Human-AI Interaction

ACM Reference Format:

Ruixuan Sun, Xinyi Li, Avinash Akella, and Joseph A. Konstan. 2025. Multi-Prompting Scenario-based Movie Recommendation with Large Language

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CHI EA '25, April 26–May 01, 2025, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3706682>

Models: Real User Case Study. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 12 pages. <https://doi.org/10.1145/3706599.3706682>

1 Introduction

Large language models (LLMs) have been developed extensively and applied to various domains to assist with a range of human tasks [6, 10, 30]. In recommender systems (RecSys), LLMs have been integrated in two main ways: 1) Enhancing RecSys training process through testing fine-tuning techniques for candidate generation [60], generating latent space or item descriptions [2, 46, 55], and designing personalized prompts for downstream recommendation tasks [17]; 2) Simulating user agents [22] or directly evaluate recommendation results [14, 54].

However, there is a lack of studies on real user evaluations of LLM-based recommenders, which is essential to assess the practical performance of RecSys [26]. LLMs are well suited to improve the transparency of recommendations due to their interactive nature [53], allowing deeper insights into human-centric values such as interpretability and trustworthiness [5]. This study addresses this gap by evaluating recommendations from pre-trained open-source LLMs in personalized scenarios. We conducted an online field experiment with hundreds of users from a movie recommender system using a chatbot interface. Through surveys and conversation analysis, we explore three key questions:

RQ1: How do users perceive the LLM recommender compared to a classic recommender experience?

RQ2: How do prompts, scenarios, or consumption factors affect perceived recommendation quality?

RQ3: What are some effective strategies users can employ with the LLM to attain more satisfactory recommendations?

We outline the main contributions of this paper as follows:

- We propose a comprehensive study design to evaluate real user experiences with LLM-driven recommendations across various scenarios. We find that while LLMs excel in providing explainable and interactive recommendations, they struggle with

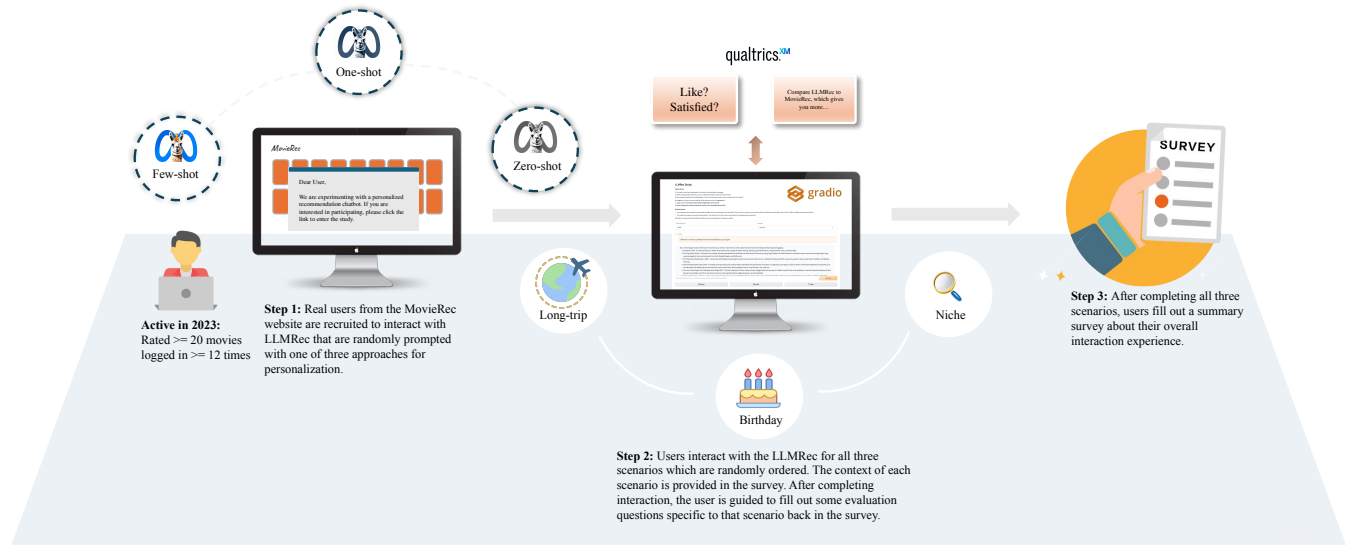


Figure 1: Overview of the LLM recommender user study design.

personalization, novelty, and diversity compared to traditional systems.

- Different prompting techniques have little effect on perceived recommendation quality, but users' consumption history plays a key role. LLMs are more effective in handling niche or unpopular movie recommendations.
- User conversation patterns indicate that providing context or examples enhances recommendation quality and user satisfaction. The results further underscore important implications for future LLM-RecSys applications.

2 Related Work

2.1 Large Language Models in Recommendation

Trained on vast textual corpora, LLMs have demonstrated remarkable capabilities on a variety of tasks relying on textual instructions [42]. Their applications extend beyond natural language processing (NLP) [41] into fields like education [4, 24] and medicine [48]. To harness the full potential of LLMs, several studies have examined prompting techniques to improve their performance. These techniques include one-shot or few-shot prompting, where the model learns from provided examples [33], and more advanced strategies such as 'Chain of Thought' (CoT) [52] and 'Tree of Thoughts' (ToT) [56], which enhance structured reasoning.

As LLMs' evolve, research has increasingly focused on their use in RecSys, reframing recommendation tasks as language-based problems and adapting LLM architectures [17, 28, 29, 61]. Another emerging trend is applying LLMs to RecSys tasks in a zero-shot setting. Liu et al. and Dai et al. examined ChatGPT's potential for RecSys tasks [13, 32], while He et al. looked at LLMs in conversational recommendation settings [20]. Sanner et al. found that LLMs are competitive with cold-start recommenders, especially with natural language preferences, and that few-shot prompting outperforms

zero-shot [44]. However, these studies mainly focused on offline evaluations, using LLMs solely as recommendation engines while neglecting online user experiences. Building on previous offline RecSys studies like [31], we conduct live experiments with users of an online movie recommendation platform. Our research explores the effects of zero-shot, one-shot, and few-shot prompting on user experiences, offering insights into engagement and LLM recommendation performance in real-world settings.

2.2 RecSys User Experience

User experience studies in RecSys are essential for developing solutions that align with user needs and expectations. Researchers have introduced human-centric frameworks to evaluate RecSys user experiences [25, 40], explored how machine learning models shape user mental models in streaming media [3], and tested how different explanations improve user trust and satisfaction. Additionally, studies have examined interactions with chatbot recommenders using PageRank algorithms [37]. However, none of these specifically addressed user experiences with LLM-based RecSys.

Granada et al. assessed an LLM-based conversational RecSys for personalized and non-personalized recommendation tasks, evaluating the associated accuracy, safety, and fairness [18]. Silva et al. leveraged ChatGPT to generate personalized movie recommendation explanation and assessed user perception of personalization, effectiveness, and persuasiveness [46]. Instead of relying on predefined candidate lists, our study focuses on user experiences with recommendations generated directly by the LLM. Along with evaluating the impact of personalized and non-personalized recommendations, we examine how different prompting techniques affect user experiences and compare LLM-based recommendations with a baseline RecSys via user ratings of the diversity, novelty, serendipity, trustworthiness, and explainability of recommendations. This helps understand how to best integrate LLMs into existing RecSys to improve user experiences.

3 Study Design

Our study follows a three-phase design, as shown in Fig. 1: 1) participant recruitment and personalized prompt generation, 2) interactions with an LLM-based recommender across different scenarios, and 3) scenario-based and summary surveys using various evaluation metrics.

3.1 Participants and Personalized Prompts

We collaborated with the movie recommendation platform *MovieLens*¹ to recruit participants. To ensure users had adequate experience with recommender systems for baseline comparison, we selected only active users who logged in at least 12 times and rated over 20 movies in 2023. This yielded a pool of 3,031 qualified users. Recruitment messages and survey access were shown on users' homepages during the experiment period. Participation was voluntary, with no incentives, encouraging thoughtful responses. Our study received IRB approval.

For the recommendation task, we used the Llama2-7b-Chat model², an open-source conversational model by Meta [49]. To address privacy concerns, we hosted the model locally instead of via a cloud API. We used three prompting techniques varying in personalization (see Fig. 2). The zero-shot prompt included each user's top three rated genres. The one-shot prompt added three movies: one recently liked, one disliked, and one recommended by the platform's model. The few-shot prompt expanded each list to four movies, as research shows more than five examples may lead to unstable outcomes [45]. Movies were further categorized into popularity tiers based on their percentiles in the MovieLens database: high (top 20%), medium (60%-80%), and low (below 60%).

3.2 User Interface and Scenarios

For user interaction, we developed the chatbot interface using Gradio³, as shown in Appendix Fig. A1. Throughout the rest of the paper, we'll refer to this interface as **LLMRec**. Users were encouraged to interact with the chatbot for as long as they needed in each scenario. To mitigate potential latency issues arising from multiple users making inference requests, we ran pilot tests to ensure the model could handle real-time inference for at least two users at once. Additionally, we included a disclaimer message and an estimated wait time on the chatbot interface to keep users informed about potential generation queues during their interactions.

Upon entering the survey, users were greeted with an informational page followed by three common movie recommendation scenarios: 1) *Ask-for-others – Friend's or Family's Birthday*; 2) *Personalization – Long Trip*; and 3) *Unpopular – Something Niche*. Detailed description of each scenario is included in Fig. 3. To mitigate potential position bias, the order of scenarios was randomized for each user within the questionnaire. This ensured that users' prior experiences did not influence their responses to subsequent scenarios.

3.3 Evaluation Metrics

After finishing their conversation with LLMRec in each scenario, users were redirected to a survey to answer two sets of questions: 1)

their overall appreciation of the recommendation scenario, focusing on enjoyment and satisfaction, and 2) a comparison between their experiences with MovieLens and LLMRec, evaluated on factors like *Personalization*, *Diversity*, *Novelty*, *Serendipity*, *Trustworthiness*, and *Explainability*. The exact wording of the survey questions, shown on the left side of Fig. 3, was adapted from prior studies by Ekstrand et al. and Nguyen [15, 38].

After completing the scenario evaluations, users were taken to a summary page to provide feedback on their overall interaction with LLMRec. Inspired by chatbot survey designs in Chaves and Gerosa's work, users rated aspects of the chatbot interaction such as Responsiveness, Understandability, Helpfulness, and whether they would like to see LLMRec integrated into the MovieLens platform in the future [11]. The survey concluded with an optional free-response section, allowing users to share additional unstructured feedback.

4 Results

Our experiment was conducted over one month from 02/13/2024 until 03/14/2024. A total of 449 users enrolled in the study (i.e. opening the questionnaire), and 178 users submitted responses. For our analysis, we focused on 160 unique users who completed all quantitative questions, excluding the optional free response. The distribution across prompting techniques was relatively balanced: 56 users in the zero-shot group, 48 in the one-shot group, and 56 in the few-shot group. The median time for completing the evaluation of all three scenarios was 17.62 minutes.

4.1 User-assessed LLMRec Quality

Based on user survey responses comparing their MovieLens to LLMRec experiences, we categorized responses into three groups: *prefer MovieLens*, *About the Same*, and *prefer LLMRec*. In comparison to MovieLens, LLMRec stood out for its explainability, but users generally found MovieLens to offer more novel, diverse, accurate, and trustworthy recommendations across all three scenarios (Fig. 4).

To gain a deeper understanding of individual user experiences, we conducted a qualitative analysis of the free-response data (N=137) using the classic grounded theory method (GMT) [8]. In this approach, three researchers who were trained in GMT independently coded the user responses. To ensure accuracy, we cross-checked 10% of each other's coding at random. Subsequently, all researchers met to do thematic analysis [12], assigning each code to one or more topic clusters and iterating this process until all ambiguities were resolved. Ultimately, we identified three major themes.

4.1.1 Defect of Recommendation Quality. The first major theme we identified centers on the attributes where LLMRec underperformed. Many users expressed dissatisfaction with the lack of personalization (N=39) and novelty (N=56) in the recommendations. Participant P50 noted, "I received repetitive recommendations for movies I had already watched. While it correctly grasped the concept of gore and violent films, it predominantly suggested love stories for me and my wife, which lacked the variety I anticipated." Another significant issue users noted was low diversity (N=10) and high-popularity (N=18). P376 summarized this by saying, "It pretty much always stuck to only the most popular movies of all time."

¹<https://movielens.org/>

²<https://huggingface.co/meta-llama/Llama-2-7b-chat-hf>

³<https://www.gradio.app/>

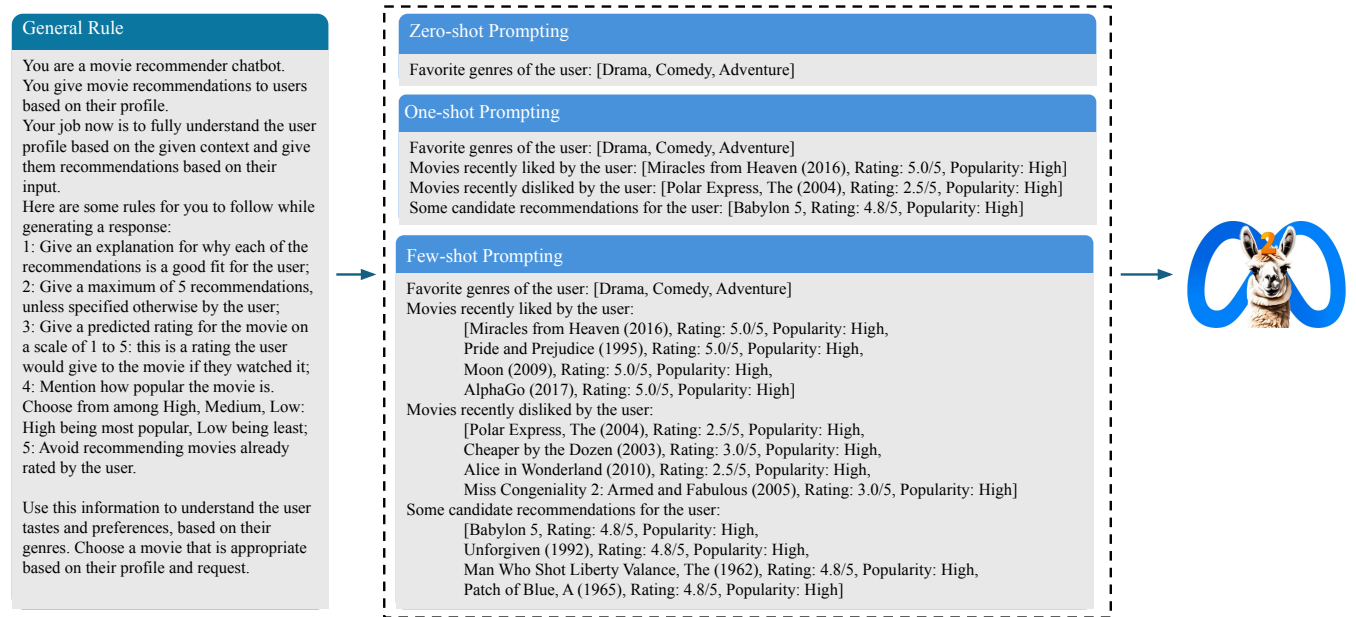


Figure 2: Diagram depicting the general rule and randomly selected prompting technique of the LLM for engaging with users at the 1st phrase of the user study.

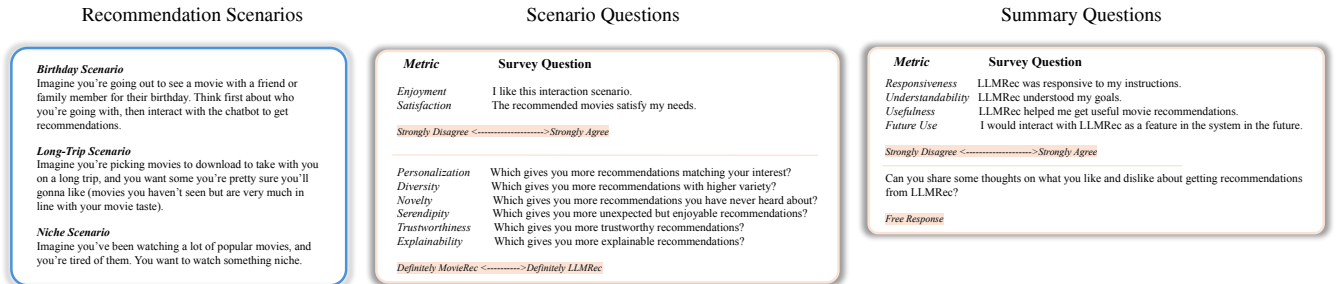


Figure 3: Left: Three recommendation scenarios. Mid: Scenario Questions; Right: Summary Questions.

4.1.2 Interactivity, Explainability, and Context. In this theme, users shared their positive interaction experience with LLMRec (N=49). P153 praised, "I like that it is interactive and takes many of my already established preferences into consideration when making recommendations." Building on this interactivity, users like P368 also enjoyed getting explanations on why they got certain recommendations, "I like that it explains why I might like the films—this is far better than the traditional MovieLens interface." In addition to explanations, users found LLMRec supported context-oriented recommendation, accommodating mood and interests of others, or even quickly extracting preferences from new users. As P407 remarked, "I like the additional option to tell the chatbot some information about what I want to watch which I can't do if I only rate movies. Additionally it can be helpful to get recommendations faster, if you're new to recommender services and haven't rated that much yet."

4.1.3 Control, Trust, and Transparency. Despite generally positive interactions, users expressed concerns about the reliability of LLMRec (N=42). One issue was the difficulty in user control. As P169 shared, "In the road trip scenario, it got fixated on horror movies and started only suggesting those, even though I dislike them very much." Similar to previous empirical study on LLMs [21], we also observed hallucination in recommendation scenarios, which deeply undermines user trust. P47 commented, "I am also wary about the chatbot suggestions because I know it can often hallucinate and not understand me clearly etc. I feel like I can't rely fully on chatbot, compared to relying on my own judgement and judgement of other users that in regular movie selection experience." This sense of disappointment prompted users to call for more transparency in LLM recommendations. P23 noted, "It was also a black box, I like knowing that the algorithm is SVD based (for example), as opposed to whatever the collaborative intelligence model has scraped from the internet."

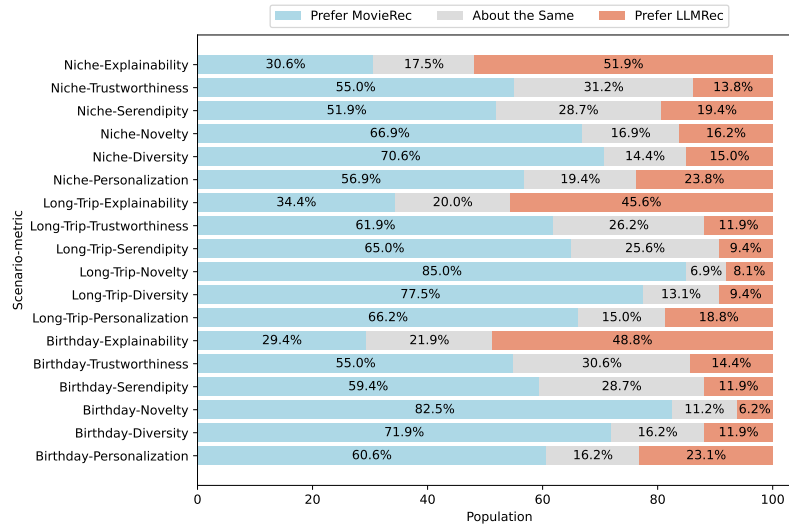


Figure 4: User perception of recommendation quality from LLMRec compared to the classic MovieLens. All LLMRec-MovieLens pairs are tested with paired t-test and shows $p \leq 0.05$ statistical significance except for the Birthday- and Long-Trip-Explainability.

With MovieLens-LLMRec comparison and qualitative analysis, we address **RQ1**: Users found LLM recommenders superior to traditional systems in explaining recommended movies. However, they expressed concerns about a lack of algorithmic transparency in the recommendation process. While users appreciated the LLM’s smooth interaction and adaptability to customized contexts, the recommendations often felt non-personalized and skewed toward popular, homogeneous content. This led to diminished user trust and reduced perceived reliability of the LLM recommender.

4.2 Effect of Prompt, Scenario, and Rating

In the second stage of our analysis, we conducted both between-subject prompt-wise tests and within-subject scenario-wise tests to examine differences across various response groups. Using one-way ANOVA and pairwise Tukey’s HSD tests [1], we assessed statistical significance. Surprisingly, we found no significant differences in evaluation metrics between different prompts (see Appendix Table A1). However, our ANOVA tests revealed significant group-wise differences in the means for *Enjoyment*, *Satisfaction*, *Novelty*, and *Serendipity* scores (Fig. 5 and Table 1). Notably, users reported greater enjoyment and satisfaction in the *Niche* scenario compared to the *Trip* scenario, and they perceived more novel and serendipitous recommendations in the *Niche* scenario than in the others.

The lack of differences across prompting techniques prompted us to explore other factors influencing user perceptions beyond LLM attributes. We categorized users into three groups based on their total number of ratings, as shown in Table 2. This classification enabled us to analyze whether a user’s historical movie consumption correlated with variations in recommendation and chatbot quality metrics (Fig. 5 and Table 3).

ANOVA revealed significant differences among groups for metrics such as recommendation *Enjoyment*, *Satisfaction*, *Personalization*, *Novelty*, and *Serendipity*, as well as chatbot *Responsiveness*, *Understandability*, and *Usefulness*. Light raters found LLMRec more enjoyable and novel than medium and heavy raters. Conversely, heavy raters reported lower satisfaction, feeling LLMRec was less personalized, responsive, and less likely to generate unexpected recommendations. They also felt it did not understand their needs as effectively as for users with fewer ratings.

Based on our findings, we answer **RQ2**: We found no significant differences in user perceptions based on varying personalization prompts. However, among the three scenarios, users reported greater enjoyment and felt their needs were better met when seeking niche recommendations, which also yielded more novel and unexpected movie suggestions. Additionally, users’ past movie consumption significantly impacts satisfaction with LLM and Chatbot recommendations. Those who rated more movies found the recommendations less personalized, leading to a more negative view of the LLMRec’s responsiveness and usefulness.

4.3 Useful Conversation Strategies

In the third analysis, we investigated user conversation data with LLMRec to explore patterns related to user experiences. Using Ordinal Linear Regression (OLR), we found significant negative correlations between the number of sentences exchanged and both Satisfaction ($coef = -0.079$, $p = 0.041$) and Understandability ($coef = -0.118$, $p = 0.002$). This suggests that users with shorter dialogues felt LLMRec better understood their goals.

4.3.1 Pattern Coding. Building on this initial finding, we conducted a qualitative analysis to identify conversation patterns associated with user responses to the *Satisfaction* question. We categorized

	F-val	P-val	B-T	B-N	T-N
<i>Enjoyment</i>	3.244	.040	-	-	.363 (.031)
<i>Satisfaction</i>	4.239	.015	-	-	.419 (.011)
<i>Novelty</i>	8.461	.000	-	.410 (.002)	.438 (.001)
<i>Serendipity</i>	3.476	.032	-	-	.319 (.027)

Table 1: ANOVA and pairwise Tukey’s HSD test results for different scenario metrics. Pairwise stats include the mean difference and p-val in bracket. N: Niche, T: Trip, B: Birthday.

	percentile	# ratings	# users
<i>light-rater</i>	<=25%	<=406	40
<i>medium-rater</i>	25 -75%	406-1522	80
<i>heavy-rater</i>	>=75%	>=1522	40

Table 2: User rating buckets

	F-val	P-val	L-M	M-H	L-H
<i>Enjoyment</i>	7.360	.001	-.429(.007)	-	.600(.001)
<i>Satisfaction</i>	8.195	.000	-	-.417(.011)	-.658(.000)
<i>Personalization</i>	6.976	.001	-	-.350(.037)	-.609(.001)
<i>Novelty</i>	7.718	.001	-.396(.003)	-	.508(.001)
<i>Serendipity</i>	9.087	.000	-	-.363(.009)	.592(.000)
<i>Responsiveness</i>	5.290	.006	-	-.538(.047)	-.825(.005)
<i>Understandability</i>	4.626	.011	-	-	-.750(.010)
<i>Usefulness</i>	5.349	.006	-	-.650(.018)	-.825(.008)

Table 3: ANOVA and pairwise Tukey’s HSD test results for different rater metrics. Pairwise stats include the mean difference and p-val in bracket. L: Light, M: Medium, H: Heavy.

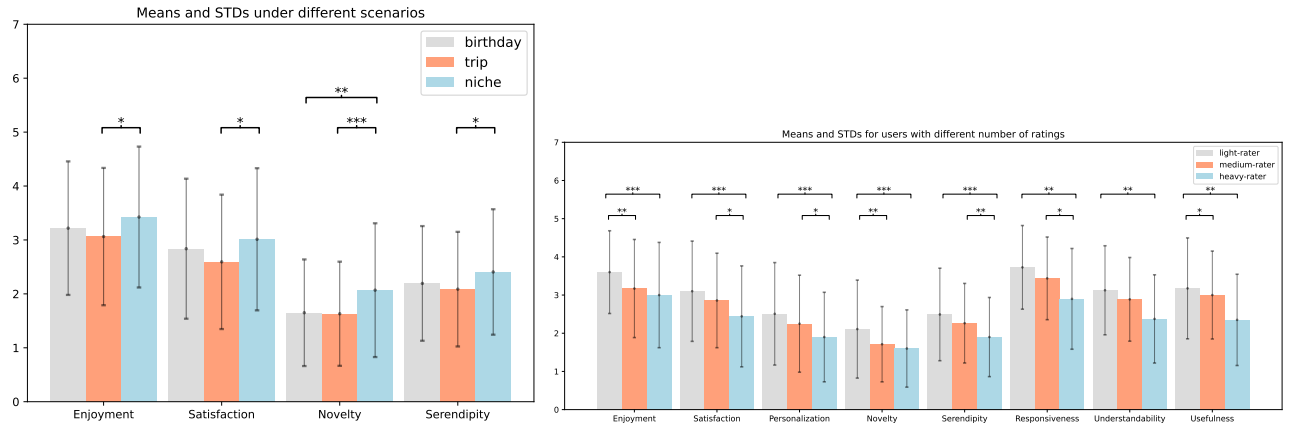


Figure 5: Within-subject and between-subject test results. Left: Differences of recommendation quality from different scenarios. Right: Differences of recommendation quality for users based on their historic rating count. Asterisk (*) indicates p-val between conditions with Tukey’s HSD test: $p < .1$ (*); $p < .05$ (); $p < .01$ (***).**

responses into positive (i.e., "Agree" and "Strongly agree") and negative (i.e., "Disagree" and "Strongly disagree"). Using GMT and

thematic analysis, we identified 10 key patterns that correlated with both positive and negative user conversations.

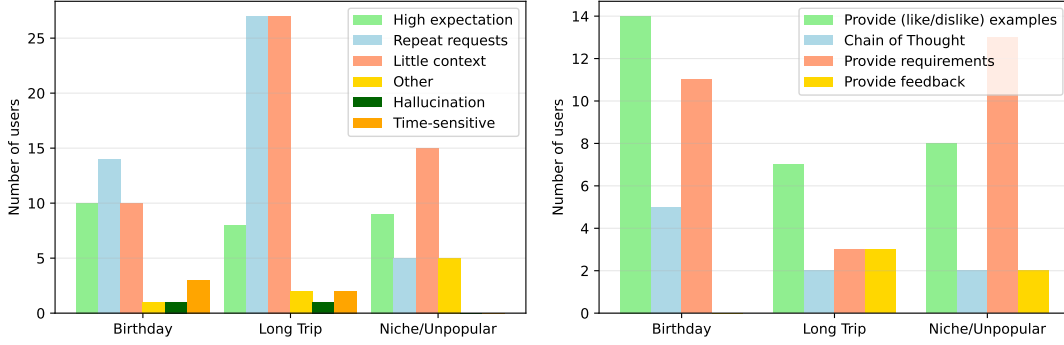


Figure 6: User conversation patterns based on qualitative coding. Left: the distribution of patterns in negative conversation. Right: the distribution of patterns in positive conversation.

Figure 6 shows that users who provided specific information, such as previously watched movies, expressed preferences (likes or dislikes), or specified genres, tend to have higher satisfaction. However, users who offered minimal context and expected LLM-Rec to infer all their preferences, treating it like a search engine, reported lower satisfaction. Based on these insights, we refined the conversation patterns and defined five discrete tags to characterize conversation attributes, shown in Table 4. Examples of these categorized conversations can be found in Appendix Table A2.

4.3.2 Regression Model. We analyzed all 417 user conversation queries across all scenarios. A researcher manually labeled conversations as "Applicable" or "Not Applicable" under each of the five tags. To ensure validation, two other researchers independently labeled 20% of the conversation data. Inter-rater reliability was assessed using Cohen's Kappa and Krippendorff's Alpha coefficients [19], both yielding values of 0.652, indicating substantial agreement among raters.

We built OLR models to predict user survey evaluation metrics based on five conversation tags, as summarized in Table 5. A key finding was that the main factor influencing user satisfaction with recommendations was the context provided by users. As shown in Fig. 6, users who were dissatisfied with LLMRec typically shared less context than those who were satisfied and offered specific examples and requirements. The regression results also indicated that providing context enhances LLMs' ability to generate more personalized and diverse recommendations. Additionally, context positively influenced LLMRec's responsiveness, understandability, and overall helpfulness. Conversely, *Dialogue* negatively impacted user-perceived understandability, suggesting that even when users communicated in plain, conversational language, they often felt the bot didn't fully grasp their objectives.

Users who attempted to test LLMRec generally reported lower satisfaction with their interactions. However, since their goal was not to obtain movie recommendations, the negative impact of the *Testing* label on recommendation or general interaction metrics is not a significant concern. Additionally, a positive relationship was found between the *Retry* label and recommendation explainability, suggesting that repeating the same queries might enhance users' understanding of why specific recommendations were made.

However, Fig. 6 indicates that repeating the same request is one of the top patterns associated with negative interaction experiences, whereas employing a chain-of-thought strategy could lead to higher satisfaction with the recommendations.

Based on our findings, we address **RQ3**: Users should provide sufficient context about their preferences when requesting recommendations from LLMs. Offering examples of likes or dislikes helps the LLM better understand user preferences. Engaging in human-like dialogue may not improve understanding, and treating LLMRec like a search engine without feedback is not ideal for quality recommendations. Repeating queries might enhance explainability temporarily, but it doesn't significantly boost overall satisfaction. Many satisfied users offered feedback or employed a chain-of-thought approach. Additionally, users should recognize the limitations of LLM recommenders and stay focused on relevant topics to avoid confusing the bot.

5 Discussion

In this section, we summarize our novel findings and future design implications for LLM recommenders from a user experience perspective in two folds: 1) How researchers can optimize LLM modeling, prompting, and information retrieval strategies to better meet user needs; and 2) How users can be better informed and educated to enhance their interaction experience with LLM recommenders, without needing technical expertise or familiarity with terminology.

5.1 Implications for LLM-Rec designers

LLMs' recommendation abilities are often constrained by training data biases toward popular content [59], but they offer two key advantages over traditional systems: 1) excellence of explaining recommendations using personal context Acharya et al., Silva et al., Wang et al., and 2) adaptation to users' real-time interest through in-context learning [7, 43]. We recommend that practitioners harness these strengths by integrating LLMs into recommendation systems to enhance user experience. However, as few-shot prompts did not significantly improve recommendations compared to zero-shot or one-shot approaches [13], anchoring LLMs in genuine item databases with techniques like retrieval-augmented generation (RAG) can be considered [27]. While RAG can effectively

Tag	Description
Dialogue	When users have a conversation by sharing details and giving feedback on how they feel about the recommendations. It should also be like talking to humans instead of a search engine, not always dumping questions or steering the response to what they want.
Context	When users clearly specify their tastes/requirements/preferences and what they're looking for. Specifically, they need to come up with something on their own, not just reuse the scenario context we provided in the survey.
Steering	When users try to make the recommendations converge to their choice without clarifying their objective. They may have multiple rounds of conversation and iteratively try to nudge the chatbot to get closer to their goal.
Testing	When users try to see how the system works and usually test it negatively. Some examples include asking questions outside of movie recommendations, or trying to decode the system prompts the bot was based on, or using inappropriate language.
Retry	When users re-generate a response to their previous query. This can be identified by repeated queries in conversation history.

Table 4: Main tags associated with user conversation with LLMRec.

	Satisfaction	Personalization	Diversity	Explainability	Responsiveness	Understandability	Helpfulness	Future Use
Dialogue	-.375(.086)	.031(.890)	-.051(.821)	.040(.854)	-.269(.240)	-.427(.050)	-.098(.665)	-.064(.775)
Context	.464(.029)	.584(.007)	.748(.001)	.210(.317)	.564(.011)	.465(.027)	.513(.017)	.188(.382)
Steering	.056(.793)	-.127(.560)	.003(.991)	.078(.715)	-.124(.575)	-.281(.188)	.162(.448)	.186(.382)
Testing	-.938(.008)	-1.061(.010)	-.816(.056)	-.778(.028)	.301(.404)	-.645(.057)	-.456(.208)	-1.026(.003)
Retry	-.021(.960)	.428(.299)	.598(.119)	.845(.041)	-.471(.212)	-.011(.977)	.201(.608)	.620(.118)

Table 5: OLR Analysis of conversation label features to user survey response metrics. Each column of metrics is run with one single OLR model. Value in each cell is formatted as coef (p-val).

re-rank more diverse though potentially less relevant recommendations [9, 14], its success depends on careful selection of embedding and retrieval models, and requires field tests to confirm its effectiveness.

The qualitative findings from surveys identified several opportunities to enhance LLM-based recommenders. Users suggested incorporating multimodal recommendations, such as movie posters or trailers, to create a more intuitive experience, potentially through visual-enhanced chatbots [57]. They also expressed a desire for improved short-term memory in LLMRec, enabling it to retain context throughout conversations, akin to MemGPT [39]. Furthermore, users wanted LLMRec to be more proactive in seeking context and feedback. Future research could investigate various LLM personas [23, 36] to achieve an optimal balance of proactivity and user-customizable recommendations.

Another concern was the balance between AI-generated content censorship and recommendation quality. Users suggested using confidence thresholds and disclaimers to address LLM hallucinations and misinformation [16, 34, 35]. However, some users, like P241, raised concerns about overly strict content guardrails: "...the guardrails were a bit too strict. I wanted to ask the system for a more naughty movie, but was prevented as it would not recommend anything with violence or sex. I can see why this is a problem in the US, but it is also a problem if the chatbot is unable to recommend high-quality movies just because they have a violent or sexual theme." This highlights a research gap for future studies to explore the right balance of AI ethics in LLM recommenders.

5.2 Learnings for End Users

The success of LLM recommenders depends on well-informed users with clear goals and smooth interaction flows. Users should be reminded to provide enough context for more personalized recommendations [47, 50], as it is difficult for designers to include the most relevant context in system prompts since user needs frequently change. Future research could explore ways to encourage users to provide more context to enhance in-context learning [7, 43], such as offering query examples or developing proactive LLMs that request more context. However, these models must balance gathering useful context with avoiding sensitive information, while keeping conversation length short to suit user impatience.

Another key implication is to inform users about LLM recommenders' capabilities and set realistic expectations [58]. High expectations can lead to dissatisfaction (see Fig. 6), like P54 shared: "While the chatbot claims to know my profile, it clearly does not. It claimed that I recently disliked the Matrix, that is not true, I liked it, and it was a long time ago. But was right about the other movies." This highlights the need for stricter rules to prevent LLM hallucinations and for educating users about LLM limitations. Encouraging users to provide corrective feedback and teaching them chain-of-thought strategies could enhance step-wise reasoning and iteratively improve recommendations.

We also see future potential in developing built-in mechanisms for LLMs to detect when users deviate from the topic and gently guide them back. Results from Fig. 6 and Table 5 suggest that discussing irrelevant topics or testing the LLM recommender leads to negative user experiences. To prevent users from unintentionally straying from the recommendation theme, LLMs could issue subtle reminders, either directly through conversation or via a UI alert.

6 Limitations and Conclusion

This paper investigates the potential of using large language models (LLMs) for personalized movie recommendations in an online field experiment. We assess the performance of LLM recommenders using a combination of between-subject prompts, historical consumption patterns, and within-subject recommendation scenarios. Analyzing conversation and survey data from 160 active users, we find that while LLMs excel in providing explainable recommendations, they lack in personalization, diversity, and user trust. Interestingly, personalized prompting techniques do not significantly affect user-perceived recommendation quality, while the number of movies a user has watched plays a more significant role. Furthermore, LLMs demonstrate a stronger ability to recommend lesser-known or niche movies. Through qualitative analysis, we identify key conversational patterns linked to positive and negative user interaction experiences and conclude that providing personal context and examples is crucial for obtaining high-quality recommendations from LLMs. As we only conducted our experiment with one LLM model (Llama2-7b-Chat), future studies should consider expanding the evaluation to other comparable or more advanced generative-AI models. We believe our findings and insights offer practical implications for improving LLM-based RecSys in real-world applications.

Acknowledgments

This work was supported by the National Science Foundation under grant IIS-2232551. We thank the MovieLens users for their participation in this study.

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A Appendices

LLMRec Study

Instructions:

- 1: You need to enter your Experiment ID as shown in the MovieRec homepage.
- 2: Select the appropriate scenario you are currently exploring from the drop-down below.
- 3: Start using the chatbot: Say something like, "Can you recommend some movies to watch with my friends?"

We suggest you reuse the current tab for all the scenarios, and it is **important** to:

- 1: **Clear** your conversation history before beginning a new scenario.
- 2: **Do not change the scenario drop-down until you are completely done with it**

Good to know:

- Your request may be queued to accomodate multiple users, so we request you to be patient. If you're part of a queue, your position will be indicated at the top-right corner of the "LLMRec" chatbot window shown below.
- The model may output erroneous text sometimes. You may ask it to re-try or ask a new question. We appreciate your patience.

Feel free to converse with the chatbot and tailor your recommendations. We hope you enjoy!

Experiment ID

10292

Scenario

Long Trip

LLMRec

I will have a six-hour trip. What movie recommendations can you give?

Sure, I'd be happy to recommend some movies for your six-hour trip! Here are a few options that are known for being entertaining and engaging:

1. Inception (2010) - A mind-bending sci-fi thriller that explores the concept of dream-sharing. Starring Leonardo DiCaprio, Joseph Gordon-Levitt, and Ellen Page.
2. The Lego Movie (2014) - A hilarious and visually stunning animated film that follows the adventures of Emmet, a young Lego builder who finds himself on a quest to stop an evil tyrant from gluing the Lego universe together. Voice cast includes Chris Pratt, Elizabeth Banks, and Will Arnett.
3. The Shawshank Redemption (1994) - A powerful and thought-provoking drama about two prisoners who form an unlikely friendship and find a way to escape from prison. Starring Tim Robbins and Morgan Freeman.
4. The Grand Budapest Hotel (2014) - A visually stunning and quirky comedy-drama that follows the adventures of Gustave H, a legendary concierge at a famous hotel in the fictional Republic of Zubrowka, and Zero Moustafa, the lobby boy who becomes his most trusted friend. Starring Ralph Fiennes, Tony Revolori, and Jude Law.
5. The Lord of the Rings: The Fellowship of the Ring (2001) - The first installment of Peter Jackson's epic trilogy follows the journey of a hobbit named Frodo as he embarks on a perilous quest to destroy the One Ring and save Middle-earth from the dark lord Sauron. Starring Elijah Wood, Viggo Mortensen, and Ian McKellen.
6. Bridesmaids (2011) - A hilarious and heartwarming comedy about a group of bridesmaids who clash as they prepare for their friend's wedding. Starring Kristen Wiig, Maya Rudolph, and Melissa McCarthy.

Say something like: Can you recommend some movies to watch with my friends?

Submit

Retry

Undo

Clear

Figure A1: Example Chatbot Recommendation Interface.

	F-val	P-val	F-O	F-Z	O-Z
Enjoyment	.208	.812	.086(.825)	.071(.867)	-.015(.994)
Satisfaction	1.213	.298	-.197(.375)	-.191(.372)	.007(.999)
Personalization	1.030	.358	.095(.791)	-.113(.699)	-.208(.327)
Diversity	2.325	.099	-.235(.161)	-.232(.146)	.003(.100)
Novelty	2.930	.054	-.266(.081)	-.238(.112)	.028(.973)
Serendipity	2.002	.136	-.057(.894)	-.232(.132)	-.176(.342)
Trustworthiness	.124	.883	-.057(.893)	-.048(.916)	.009(.997)
Explainability	.008	.993	-.008(.999)	-.018(.992)	-.001(.998)
Responsiveness	1.020	.363	-.161(.772)	-.321(.329)	-.161(.772)
Understandability	.278	.758	-.161(.765)	-.125(.839)	.036(.987)
Usefulness	.928	.397	-.140(.837)	-.321(.365)	-.182(.742)
Future-Use	1.107	.333	-.095(.935)	-.375(.325)	-.280(.559)

Table A1: ANOVA and pairwise Tukey’s HSD test results for different prompt metrics. Pairwise stats include the mean difference and p-val in bracket. F: Few-shot, O: One-shot, Z: Zero-shot.

Example Raw User Conversation with LLMRec	Dialogue	Context	Steering	Testing	Retry
"Can you recommend a few movies for me?"	NA	NA	NA	NA	NA
"Can you recommend a movie for a date. it has to be something i like but she has to like it too. shes a bit more normal then me', 'I dont want a rom com', 'something i havent seen', 'those are too weird for a first date with someone whos taste i dont know. also ive seen 3 of those. try again. Ideally it would be a movie that maves the viewer feel sophisticated', '2 and 3 are good recs thanks.'	A	A	A	NA	NA
'Can you recommend some movie for me', 'i have watched inception', 'i have already watched the matrix', 'Can you recommend some movie for me, 'but i have already watched inception', 'Can you recommend some movie for me', 'but i have already watched inception, the shawshank redemption and the lords of the rings'	NA	NA	A	NA	A
"Ok, that seems useless and tedious, but let's try this. Recommend 10 movies", "I've seen all of those. Recommend 10 more movies and don't recommend those again. Also, I hated The Godfather (1972) and The Big Lebowski (1998)", "I've seen all of those. Recommend 10 more movies I haven't seen. I liked all of them except The Lobster (2015)", "Who directed The Lighthouse (2019)?", 'List all movies directed by Robert Eggers', "That isn't correct. Robert Eggers also directed The Northman and The Witch", 'I've watched Eternal Sunshine of the Spotless Mind (2004), The Lighthouse (2019), The Florida Project (2017), The Killing of a Sacred Deer (2017), The Witch (2015), The Invitation (2015), The Love Witch (2016), and The Northman (2022). Recommend 10 more movies I haven't seen. Don't recommend horror films.", 'Stop', "Recommend 10 movies I haven't seen that my boyfriend might also like", 'I asked for 10 and you give me 5', "I can't find any information about The Prisoner (1977)"	A	A	A	A	A
'can you recommend some movies that are perfect for a road trip? preferably something with action, "recomend me a movie that I haven't seen", 'something like "shutter island"', "can you recommend me a good oscar worthy movie preferably something I haven't seen", 'can you give me list', 'can you recommend me a list of good Thiller-worthy movies preferably something I haven't seen"	NA	A	A	NA	NA

Table A2: Example user conversations and their associated taggings. A denotes Applicable, NA denotes Not Applicable.